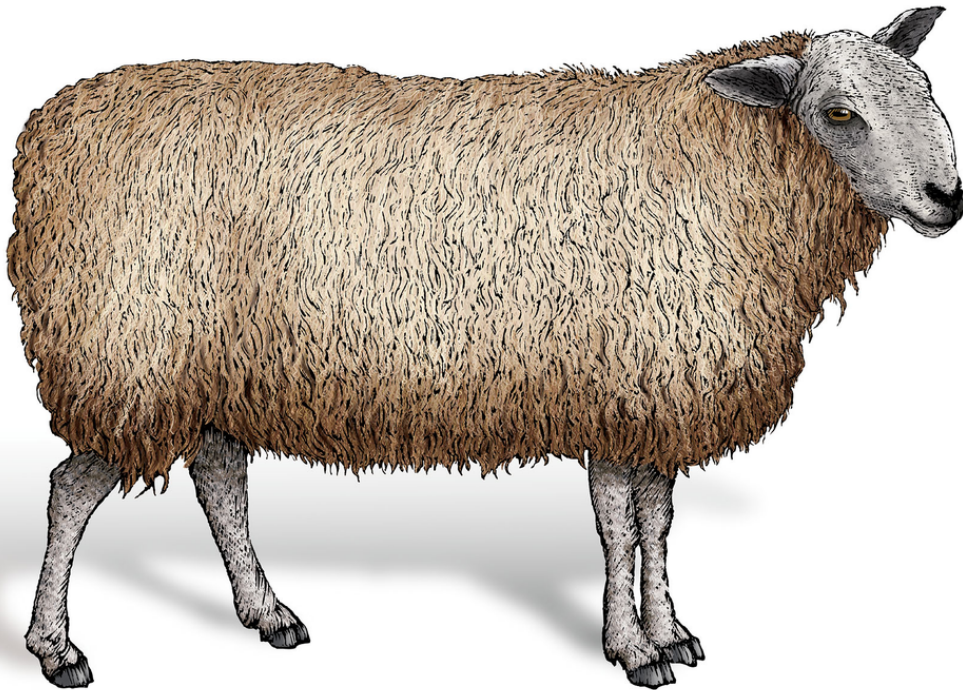


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Practical Time Series Analysis

Prediction with Statistics & Machine Learning



Aileen Nielsen

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Prediction with Statistics and
Machine Learning

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Practical Time Series Analysis

by Aileen Nielsen

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Preface

Weather, stock markets, and heartbeats. They all form time series. If you're interested in diverse data and forecasting the future, you're interested in time series analysis.

Welcome to *Practical Time Series Analysis*! If you picked this book up, you've probably already noticed that time series data is everywhere. Time series data grows increasingly ubiquitous and important as the big data ecosystem expands. For better or worse, sensors and tracking mechanisms are everywhere, and as a result there are unprecedented amounts of high-quality time series data available. Time series are uniquely interesting because they can address questions of causality, trends, and the likelihood of future outcomes. This book will take you through the major techniques that are most commonly applied to time series to address such questions.

Time series data spans a wide range of disciplines and use cases. It can be anything from customer purchase histories to conductance measurements of a nano-electronic system to digital recordings of human language. One point we discuss throughout the book is that time series analysis applies to a surprisingly diverse set of data. Any data that has an ordered axis can be analyzed with time series methods, even if that ordered axis is not time per se. Traditional time series data, such as stock data and weather patterns, can be analyzed with time series methods, but so can quirky data sets such as spectrographs of wine, where the "time" axis is actually an axis of frequency. Time series are everywhere.

Who Should Read This Book

There are two kinds of intended readers for this book. The first and larger category of reader is that of a data scientist who has rarely worked with time series data. This person could be an industry veteran or a junior analyst. The more experienced data analyst can skim the introductory conceptual areas of each chapter but will still benefit from this book's discussions about best practices as well as pitfalls of

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working with time series data. A newer data analyst might consider working through the book in its entirety, although I have tried to keep each topic as self-contained as possible.

The second category of reader is someone supervising analytics at an organization with an extensive in-house data collection. If you are a member of this group, you will still need some technical background, but it's not necessary that you be currently coding in your professional life. For such a reader, this book is useful to point out opportunities for your organization to use time series analysis even if it is not currently practiced in-house. This book will point you to new kinds of questions and analyses your organization can address with your existing data resources.

Expected Background

With respect to coding, you should have some familiarity with R and Python, especially with certain fundamental packages (in Python: NumPy, Pandas, and `scikit-learn`; and in R: `data.table`). The code samples should be readable even without all the background, but in that case you may need to take a short detour to familiarize yourself with these packages. This is most likely the case with respect to R's `data.table`, an underused but highly performant data frame package that has fantastic time functionality.

In all cases, I have provided brief overviews of the related packages, some example code, and descriptions of what the code does. I also point the reader toward more complete overviews of the most used packages.

With respect to statistics and machine learning, you should have some familiarity with:

Introductory statistics

Ideas such as variance, correlation, and probability distributions

Machine learning

Clustering and decision trees

Neural networks

What they are and how they are trained

For these cases, I provide a brief overview of such concepts within the text, but the uninitiated should read more deeply about them before continuing with some chapters. For most topics, I provide links to recommended free online resources for brief tutorials on the fundamentals of a given topic or technique.

Why I Wrote This Book

I wrote this book for three reasons.

First, time series is an important aspect of data analysis but one that is not found in the standard data science toolkit. This is unfortunate both because time series data is increasingly available and also because it answers questions that cross-sectional data cannot. An analyst who does not know fundamental time series analysis is not making the most of their data. I hoped that this book could fill an existing and important void.

Second, when I started writing this book, I knew of no centralized overview of the most important aspects of time series analysis from a modern data science perspective. There are many excellent resources available for traditional time series analysis, most notably in the form of classic textbooks on statistical time series analysis. There are also many excellent individual blog posts on both traditional statistical methods and on machine learning or neural network approaches to time series. However, I could not identify a single centralized resource to outline all these topics and relate them to one another. The goal of this book is to provide that resource: a broad, modern, and practical overview of time series analysis covering the full pipeline for time series data and modeling. Again, I hoped that this book could fill an existing and important void.

Third, time series is an interesting topic with quirky data concerns. Problems associated with data leakage, lookahead, and causality are particularly fun from a time series perspective, as are many techniques that apply uniquely to data ordered along some kind of time axis. Surveying these topics broadly and finding a way to catalog them was another motivation to write this book.

Navigating This Book

This book is organized roughly as follows:

History

Chapter 1 presents a history of time series forecasting, all the way from the ancient Greeks to modern times. This puts our learning in context so we can understand what a traditional and storied discipline we study in this book.

All about the data

Chapters **2**, **3**, **4**, and **5** tackle problems relating to obtaining, cleaning, simulating, and storing time series data. These chapters are tied together by their concern for all that comes before you can actually perform time series analysis. Such topics are rarely discussed in existing resources but are important in most data pipelines. Data identification and cleaning represent a large portion of the work done by most time series analysts.

Models, models, models

Chapters **6**, **7**, **8**, **9**, and **10** cover a variety of modeling techniques that can be used for time series analysis. We start with two chapters on statistical methods, covering standard statistical models, such as ARIMA and Bayesian state space models. We then apply more recently developed

methods, such as machine learning and neural network, to time series data, highlighting the challenges of data processing and data layout when time series data is used for fitting models that are not inherently time aware, such as decision trees.

Post-modeling considerations

Chapters 11 and 12 cover accuracy metrics and performance considerations, respectively, to provide some guidance as to what you should consider after you have performed your first iterations of time series modeling.

Real-world use cases

Chapters 13, 14, and 15 provide case studies from healthcare, finance, and government data, respectively.

Comments on recent happenings

Chapters 16 and 17 briefly cover recent time series developments and predictions for the future. Chapter 16 is an overview of a variety of automated time series packages, some open source and developed as an academic effort, and some coming from large tech companies. These tools are under active development as efforts ramp up to improve time series forecasting at scale via automated processes. Chapter 17 discusses some predictions for the future of time series analysis as the big data ecosystem grows and as we learn more about how big data can assist time series analysis.

In general, I recommend reading through a chapter before attempting to work with the code. There are usually a few new concepts introduced in each chapter, and it can be helpful to give these some attention before turning to your keyboard. Also, in most cases the code to execute certain models is relatively simple so that the conceptual understanding will be your main skill gained, with knowledge of the APIs of important packages being a secondary benefit that will come much more easily if you pay attention to the concepts.

Also, the book is written so that it is most sensibly read from beginning to end (with later chapters referring to concepts covered earlier in the book), but, again, I purposely kept the chapters as self-contained as possible to enable more experienced readers to jump around as convenient.

Online Resources

The [GitHub repository for this book](#) includes much of the code discussed in this book in a form where you can run it on data sets, which are also provided in the repository. In some cases, variable names, formatting, and such will not be identical to that produced in the book, but it should be easily linked (for example, in some cases variable names were shortened in the book due to formatting constraints).

Additionally, should you enjoy video presentations, I have two online tutorials covering some of the content of this book with a Python focus. If you want to complement this book with a video presentation, consider the following resources:

- **Time Series Analysis** (SciPy 2016)
- **Modern Time Series Analysis** (SciPy 2019)

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

Constant width

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

Constant width bold

Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows text that should be replaced with user-supplied values or by values determined by context.

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

This element indicates a warning or caution.

Using Code Examples

Supplemental material (code examples, exercises, etc.) is available for download at <https://github.com/PracticalTimeSeriesAnalysis/BookRepo>.

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Any mistakes or bugs you find in this book are mine and mine alone. I welcome feedback at aileen.a.nielsen@gmail.com.

Chapter 1. Time Series: An Overview and a Quick History

Time series data and its analysis are increasingly important due to the massive production of such data through, for example, the internet of things, the digitalization of healthcare, and the rise of smart cities. In the coming years we can expect the quantity, quality, and importance of time series data to grow rapidly.

As continuous monitoring and data collection become more common, the need for competent time series analysis with both statistical and machine learning techniques will increase. Indeed, the most promising new models combine both of these methodologies. For this reason, we will discuss each at length. We will study and use a broad range of time series techniques useful for analyzing and predicting human behavior, scientific phenomena, and private sector data, as all these areas offer rich arrays of time series data.

Let's start with a definition. *Time series analysis* is the endeavor of extracting meaningful summary and statistical information from points arranged in chronological order. It is done to diagnose past behavior as well as to predict future behavior. In this book we will use a variety of approaches, ranging from hundred-year-old statistical models to newly developed neural network architectures.

None of the techniques has developed in a vacuum or out of purely theoretical interest. Innovations in time series analysis result from new ways of collecting, recording, and visualizing data. Next we briefly discuss the emergence of time series analysis in a variety of applications.

The History of Time Series in Diverse Applications

Time series analysis often comes down to the question of causality: how did the past influence the future? At times, such questions (and their answers) are treated strictly within their discipline rather than as part of the general discipline of time series analysis. As a result, a variety of disciplines have contributed novel ways of thinking about time series data sets.

In this section we will survey a few historical examples of time series data and analysis in these disciplines:

- Medicine
- Weather

- Economics
- Astronomy

As we will see, the pace of development in these disciplines and the contributions originating in each field were strongly tied to the nature of the contemporaneous time series data available.

Medicine as a Time Series Problem

Medicine is a data-driven field that has contributed interesting time series analysis to human knowledge for a few centuries. Now, let's study a few examples of time series data sources in medicine and how they emerged over time.

Medicine got a surprisingly slow start to thinking about the mathematics of predicting the future, despite the fact that prognoses are an essential part of medical practice. This was the case for many reasons. Statistics and a probabilistic way of thinking about the world are recent phenomena, and these disciplines were not available for many centuries even as the practice of medicine developed. Also, most doctors practiced in isolation, without easy professional communication and without a formal recordkeeping infrastructure for patient or population health. Hence, even if physicians in earlier times had been trained as statistical thinkers, they likely wouldn't have had reasonable data from which to draw conclusions.

This is not at all to criticize early physicians but to explain why it is not too surprising that one of the early time series innovations in population health came from a seller of hats rather than from a physician. When you think about it, this makes sense: in earlier centuries, an urban hat seller would likely have had more practice in recordkeeping and the art of spotting trends than would a physician.

The innovator was John Graunt, a 17th-century London haberdasher. Graunt undertook a study of the death records that had been kept in London parishes since the early 1500s. In doing so, he originated the discipline of demography. In 1662, he published *Natural and Political Observations . . . Made upon the Bills of Mortality* (See [Figure 1-1](#)).

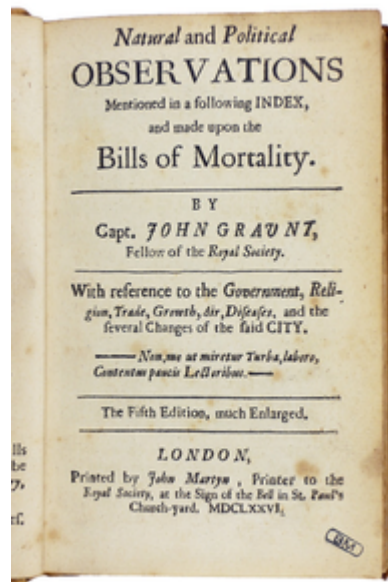


Figure 1-1. John Graunt's actuarial tables were one of the first results of time-series-style thinking applied to medical questions. Source: [Wikipedia](#).

In this book, Graunt presented the first *life tables*, which you may know as *actuarial tables*. These tables show the probability that a person of a given age will die before their next birthday. Graunt, as the first person known to have formulated and published life tables, was also the first documented statistician of human health. His life tables looked something like [Table 1-1](#), which is taken from some [Rice University statistics course notes](#).

Table 1-1. A sample of John Graunt's life tables

Age	Proportion of deaths in the interval	Proportion surviving until start of interval
0-6	0.36	1.0
7-16	0.24	0.64
17-26	0.15	0.40
27-36	0.09	0.25

Unfortunately, Graunt's way of thinking mathematically about human survival did not take. A more connected and data-driven world began to form—complete with nation states, accreditation, professional societies, scientific journals, and, much later, government-mandated health recordkeeping—but medicine continued to focus on physiology rather than statistics.

There were understandable reasons for this. First, the study of anatomy and physiology in small numbers of subjects had provided the major advances in medicine for centuries, and most humans (even scientists) hew to what works for them as long as possible. While a focus on physiology was so successful, there was no reason to look further afield. Second, there was very little reporting infrastructure in place for physicians to tabulate and share information on the scale that would make statistical methods superior to clinical observations.

Time series analysis has been even slower to come into mainstream medicine than other branches of statistics and data analysis, likely because time series analysis is more demanding of recordkeeping systems. Records must be linked together over time, and preferably collected at regular intervals. For this reason, time series as an epidemiological practice has only emerged very recently and incrementally, once sufficient governmental and scientific infrastructure was in place to ensure reasonably good and lengthy temporal records.

Likewise, individualized healthcare using time series analysis remains a young and challenging field because it can be quite difficult to create data sets that are consistent over time. Even for small case-study-based research, maintaining both contact with and participation from a group of individuals is excruciatingly difficult and expensive. When such studies are conducted for long periods of time, they tend to become canonical in their fields—and repeatedly, or even excessively researched—because their data can address important questions despite the challenges of funding and management.¹

Medical instruments

Time series analysis for individual patients has a far earlier and more successful history than that of population-level health studies. Time series analysis made its way into medicine when the first practical electrocardiograms (ECGs), which can diagnose cardiac conditions by recording the electrical signals passing through the heart, were invented in 1901 (see Figure 1-2). Another time series machine, the electroencephalogram (EEG), which noninvasively measures electrical impulses in the brain, was introduced into medicine in 1924, creating more opportunities for medical practitioners to apply time series analysis to medical diagnosis (see Figure 1-3).

Both of these time series machines were part of a larger trend of enhancing medicine with repurposed ideas and technologies coming out of the second Industrial Revolution.

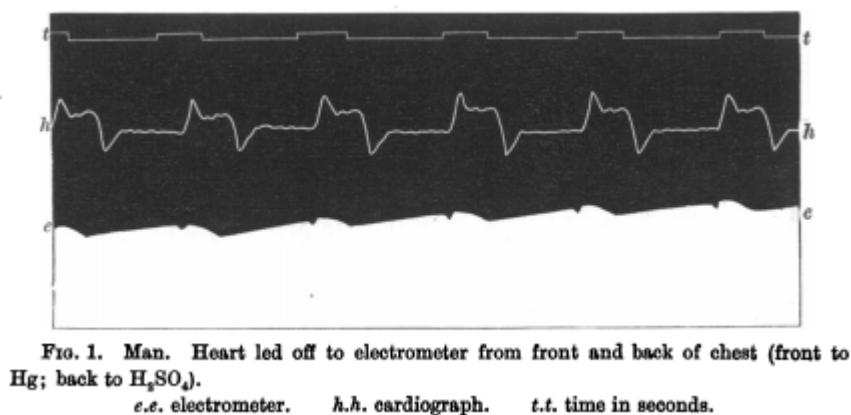


Figure 1-2. An early ECG recording from the original 1877 paper by Augustus D. Waller, M.D, “*A Demonstration on Man of Electromotive Changes Accompanying the Heart’s Beat*”. The earliest ECGs were difficult to construct and use, so it was a few more decades before they became a practical tool for physicians.

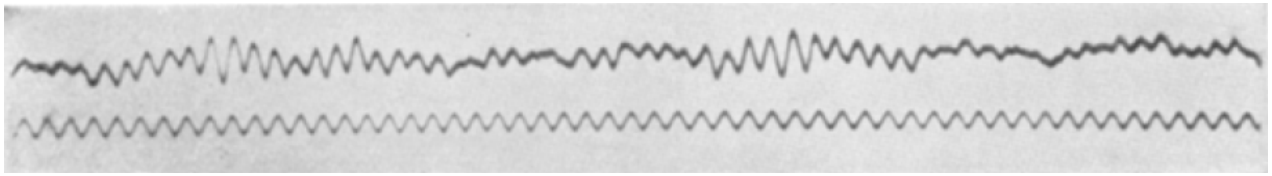


Figure 1-3. The first human EEG recording, from 1924. Source: [Wikipedia](#).

ECG and EEG time series classification tools remain active areas of research for very practical purposes, such as estimating the risk of a sudden cardiac crisis or a seizure. These measurements are rich sources of data, but one “problem” with such data is that it tends to be available only for patients with specific ailments. These machines do not generate long-range time series that can tell us more broadly about human health and behavior, as their measurements are seldom applied for long periods of time or before a disease has emerged in a patient.

Luckily, from a data analysis point of view, we are moving past the era where ECGs and the like are the dominant medical time series available. With the advent of wearable sensors and “smart” electronic medical devices, many healthy humans take routine measurements automatically or with minimal manual input, leading to the ongoing collection of good longitudinal data about both sick and healthy people. This is in stark contrast to the last century’s medical time series data, which was almost exclusively measured on sick people and which was very limited in access.

As recent news coverage has shown, a variety of nontraditional players are entering the medical field, ranging from enormous social media companies to financial institutions to retail giants.² They likely all plan to use large data sets to streamline healthcare. There aren’t just new players in the healthcare field—there are also new techniques. The personalized DNA-driven medicine means that time series *data* is increasingly measured and valued. Thanks to burgeoning modern healthcare data sets, both healthcare and time series analysis will likely evolve in the coming years, particularly in response to the lucrative data sets of the healthcare sector. Hopefully this will happen in such a way that time series can benefit everyone.

Forecasting Weather

For obvious reasons, predicting the weather has long been a preoccupation to many. The ancient Greek philosopher Aristotle delved into weather with an entire treatise (*Meteorology*), and his ideas about the causes and sequencing of the weather remained dominant until the Renaissance. At that time, scientists began to collect weather-related data with the help of newly invented instruments, such as the barometer, to measure the state of the atmosphere. They used these instruments to record time series at daily or even hourly intervals. The records were kept in a variety of locations, including private diaries and local town logbooks. For centuries this remained the only way that Western civilization tracked the weather.

Greater formalization and infrastructure for weather recording arrived in the 1850s when Robert FitzRoy was appointed the head of a new British government department to record and publish weather-related data for sailors.³ FitzRoy coined the term *weather forecast*. At the time, he was

criticized for the quality of his forecasts, but he is now regarded to have been well ahead of his time in the science he used to develop them. He established the custom of printing weather forecasts in the newspaper; they were the first forecasts printed in *The Times* of London. FitzRoy is now celebrated as the “father of forecasting.”

In the late 19th century—hundreds of years after many atmospheric measurements had come into use—the telegraph allowed for fast compilations of atmospheric conditions in time series from many different locations. This practice became standard in many parts of the world by the 1870s and led to the creation of the first meaningful data sets for predicting local weather based on what was happening in other geographic locations.

By the turn of the 20th century, the idea of forecasting the weather with computational methods was vigorously pursued with the help of these compiled data sets. Early endeavors at computing the weather required a spectacular amount of effort but gave poor results. While physicists and chemists had well-proven ideas about the relevant natural laws, there were too many natural laws to apply all at once. The resulting system of equations was so complex that it was a notable scientific breakthrough the first time someone even attempted to do the calculations.

Several decades of research followed to simplify the physical equations in a way that increased accuracy and computational efficiency. These tricks of the trade have been handed down even to current weather prediction models, which operate on a mix of known physical principles and proven heuristics.

Nowadays many governments make highly granular weather measurements from hundreds or even thousands of weather stations around the world, and these predictions are grounded in data with precise information about weather station locations and equipment. The roots of these efforts trace back to the coordinated data sets of the 1870s and even earlier to the Renaissance practice of keeping local weather diaries.

Unfortunately, weather forecasting is an example of the increasing attacks on science that reach even into the domain of time series forecasting. Not only have time series debates about global temperatures been politicized, but so have more mundane time series forecasting tasks, such as **predicting the path of a hurricane**.

Forecasting Economic Growth

Indicators of production and efficiency in markets have long provided interesting data to study from a time series analysis. Most interesting and urgent has been the question of forecasting future economic states based on the past. Such forecasts aren’t merely useful for making money—they also help promote prosperity and avert social catastrophes. Let’s discuss some important developments in the history of economic forecasting.

Economic forecasting grew out of the anxiety triggered by episodic banking crises in the United States and Europe in the late 19th and early 20th centuries. At that time, entrepreneurs and researchers alike

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drew inspiration from the idea that the economy could be likened to a cyclical system, just as the weather was thought to behave. With the right measurements, it was thought, predictions could be made and crashes averted.

Even the language of early economic forecasting mirrored the language of weather forecasting. This was unintentionally apt. In the early 20th century, economic and weather forecasting were indeed alike: both were pretty terrible. But economists' aspirations created an environment in which progress could at least be hoped for, and so a variety of public and private institutions were formed for tracking economic data. Early economic forecasting efforts led to the creation of economic indicators and tabulated, publicly available histories of those indicators that are still in use today. We will even use some of these in this book.

Nowadays, the United States and most other nations have thousands of government researchers and recordkeepers whose jobs are to record data as accurately as possible and make it available to the public (see [Figure 1-4](#)). This practice has proven invaluable to economic growth and the avoidance of economic catastrophe and painful boom and bust cycles. What's more, businesses benefit from a data-rich atmosphere, as these public data sets permit transportation providers, manufacturers, small business owners, and even farmers to anticipate likely future market conditions. This all grew out of the attempt to identify "business cycles" that were thought to be the causes of cyclical banking failures, an early form of time series analysis in economics.

<u>BUSINESS CYCLE</u>		<u>DURATION IN MONTHS</u>			
<u>REFERENCE DATES</u>					
Peak	Trough	Contraction	Expansion	Cycle	
<i>Quarterly dates are in parentheses</i>		<i>Peak to Trough</i>	<i>Previous trough to this peak</i>	<i>Trough from Previous Trough</i>	<i>Peak from Previous Peak</i>
	December 1854 (IV)	--	--	--	--
June 1857(II)	December 1858 (IV)	18	30	48	--
October 1860(III)	June 1861 (III)	8	22	30	40
April 1865(I)	December 1867 (I)	32	46	78	54
June 1869(II)	December 1870 (IV)	18	18	36	50
October 1873(III)	March 1879 (I)	65	34	99	52
March 1882(I)	May 1885 (II)	38	36	74	101
March 1887(II)	April 1888 (I)	13	22	35	60
July 1890(III)	May 1891 (II)	10	27	37	40
January 1893(I)	June 1894 (II)	17	20	37	30
December 1895(IV)	June 1897 (II)	18	18	36	35
June 1899(III)	December 1900 (IV)	18	24	42	42
September 1902(IV)	August 1904 (III)	23	21	44	39

Figure 1-4. The US federal government funds many government agencies and related nonprofits that record vital statistics as well as formulate economic indicators. Source: *National Bureau of Economic Research*.

Much of the economic data collected by the government, particularly the most newsworthy, tends to be a proxy for the population's overall economic well-being. One example of such vital information comes from the number of people requesting unemployment benefits. Examples include the government's estimates of the gross domestic product and of the total tax returns received in a given year.

Thanks to this desire for economic forecasting, the government has become a curator of data as well as a collector of taxes. The collection of this data enabled modern economics, the modern finance industry, and data science generally to blossom. Thanks to time series analysis growing out of economic questions, we now safely avert many more banking and financial crises than any government could have in past centuries. Also, hundreds of time series textbooks have been written in the form of economics textbooks devoted to understanding the rhythms of these financial indicators.

Trading markets

Let's get back to the historical side of things. As government efforts at data collection met with great success, private organizations began to copy government recordkeeping. Over time, commodities and stock exchanges became increasingly technical. Financial almanacs became popular, too. This happened both because market participants became more sophisticated and because emerging technologies enabled greater automation and new ways of competing and thinking about prices.

All this minute recordkeeping gave rise to the pursuit of making money off the markets via math rather than intuition, in a way driven entirely by statistics (and, more recently, by machine learning). Early pioneers did this mathematical work by hand, whereas current "quants" do this by very complicated and proprietary time series analytic methods.

One of the pioneers of *mechanical trading*, or time series forecasting via algorithm, was Richard Dennis. Dennis was a self-made millionaire who famously turned ordinary people, called the Turtles, into star traders by teaching them a few select rules about how and when to trade. These rules were developed in the 1970s and 1980s and mirrored the "AI" thinking of the 1980s, in which heuristics still strongly ruled the paradigm of how to build intelligent machines to work in the real world.

Since then many "mechanical" traders have adapted these rules, which as a result have become less profitable in a crowded automated market. Mechanical traders continue to grow in number and wealth, they are continually in search of the next best thing because there is so much competition.

Astronomy

Astronomy has always relied heavily on plotting objects, trajectories, and measurements over time. For this reason, astronomers are masters of time series, both for calibrating instruments and for studying their objects of interest. As an example of the long history of time series data, consider that sunspot time series were recorded in ancient China as early as 800 BC, making sunspot data collection one of the most well-recorded natural phenomena ever.

Some of the most exciting astronomy of the past century relates to time series analysis. The discovery of variable stars (which can be used to deduce galactic distances) and the observation of transitory events such as supernovae (which enhance our understanding of how the universe changes over time) are the result of monitoring live streams of time series data based on the wavelengths and intensities of light. Time series have had a fundamental impact on what we can know and measure about the universe.

Incidentally, this monitoring of astronomical images has **even allowed astronomers to catch events as they are happening** (or rather as we are able to observe them, which may take millions of years).

In the last few decades, the availability of explicitly timestamped data, as formal time series, has exploded in astronomy with a wide array of new kinds of telescopes collecting all sorts of celestial data. Some astronomers have even referred to a time series “data deluge.”

This document was truncated here because it was created in the Evaluation Mode.

